

§ 12.3 Limits & continuity

Defn $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ means: for each $\epsilon > 0$

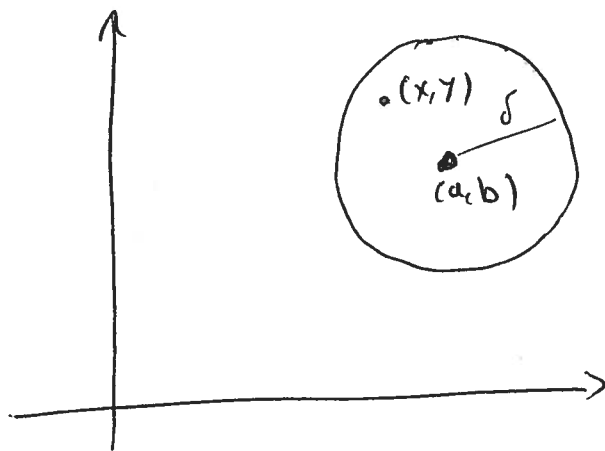
there exists a $\delta > 0$ s.t. $|f(x,y) - L| < \epsilon$
provided that $0 < \|(x,y) - (a,b)\| < \delta$

So $f(x,y)$ approaches L as (x,y) approaches (a,b)

More concisely: $\forall \epsilon > 0 \exists \delta > 0$ s.t. $0 < \|(x,y) - (a,b)\| < \delta$
 $\Rightarrow |f(x,y) - L| < \epsilon$

Note that $\|(x,y) - (a,b)\| = \sqrt{(x-a)^2 + (y-b)^2}$ so

$\|(x,y) - (a,b)\| < \delta$
is a circle with center
 (a,b) and radius δ



Note that $|x-a| \leq \|(x,y) - (a,b)\| < \delta$
 $|y-b| \leq \|(x,y) - (a,b)\| < \delta$

So (x,y) near (a,b) implies x near a & y near b

Note that $|x-a| < \delta$ & $|y-b| < \delta \Rightarrow \|(x,y) - (a,b)\| = \sqrt{(x-a)^2 + (y-b)^2} < \sqrt{2} \cdot \delta$

So x near a & y near b implies (x,y) near (a,b)

$$\varepsilon_x \quad \text{If } \lim_{x \rightarrow a} f(x) = L \quad \text{and} \quad \lim_{y \rightarrow b} g(y) = M$$

$$\text{then } \lim_{(x,y) \rightarrow (a,b)} f(x) \cdot g(y) = L \cdot M$$

Proof we know that $\forall \varepsilon_1, \varepsilon_2 > 0 \quad \exists \delta_1, \delta_2 > 0$ s.t.

$$|x - a| < \delta_1 \Rightarrow |f(x) - L| < \varepsilon_1$$

$$|y - b| < \delta_2 \Rightarrow |g(y) - M| < \varepsilon_2$$

$$\text{In particular } |f(x)| < |L| + \varepsilon_1 \quad \& \quad |g(y)| < |M| + \varepsilon_2$$

Fix $\varepsilon > 0$
We assume that $0 < \varepsilon_1, \varepsilon_2 < \min\left\{1, \frac{\varepsilon}{2(|M|+1)}, \frac{\varepsilon}{2(|L|+1)}\right\}$

$$\text{Let } \delta = \min\{\delta_1, \delta_2\} \quad \text{then}$$

$$\begin{aligned} |f(x)g(y) - LM| &= |f(x)g(y) - Lg(y) + Lg(y) - LM| \\ &\leq |(f(x) - L)g(y)| + |L(g(y) - M)| \leq \varepsilon_1(|M|+1) + |L| \cdot \varepsilon_2 \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \quad \square \end{aligned}$$

$$\underline{\varepsilon_x} \quad \text{If } \lim_{(x,y) \rightarrow (a,b)} f(x,y) = L \quad \& \quad \lim_{(x,y) \rightarrow (a,b)} g(x,y) = M$$

$$\text{then } \lim_{(x,y) \rightarrow (a,b)} (f(x,y) + g(x,y)) = L + M$$

Thm If $p(x, y)$ and $q(x, y)$ are polynomials,

then ① $\lim_{(x, y) \rightarrow (a, b)} p(x, y) = p(a, b)$

② If $q(a, b) \neq 0$ then $\lim_{(x, y) \rightarrow (a, b)} \frac{p(x, y)}{q(x, y)} = \frac{p(a, b)}{q(a, b)}$

③ If $q(a, b) = 0$ & $p(a, b) \neq 0$ then
 $\lim_{(x, y) \rightarrow (a, b)} \frac{p(x, y)}{q(x, y)}$ D.N.E.

Eg $\lim_{(x, y) \rightarrow (1, 2)} x^2 + y^2 = 1 + 4 = 5$ $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 + y^2}{x + y - 1} = \frac{0}{-1} = 0$

$\lim_{(x, y) \rightarrow (2, 2)} \frac{x^2 + y^2}{x^2 - y^2} = \frac{8}{0}$ DNE

$\lim_{(x, y) \rightarrow (0, 0)} \frac{e^{x^2 + y^2}}{\cos(x + y)} = \text{can't do this yet!}$

Thm If $g(x, y)$ is continuous at (a, b) $\left(\lim_{(x, y) \rightarrow (a, b)} g(x, y) = g(a, b) \right)$
and $f(t)$ is continuous at $t_0 = g(a, b)$ $\left(\lim_{t \rightarrow t_0} f(t) = f(t_0) \right)$
then $f \circ g$ is continuous at (a, b)
& so $\lim_{(x, y) \rightarrow (a, b)} f(g(x, y)) = f(g(a, b))$

$$\lim_{(x,y) \rightarrow (0,0)} e^{x^2+y^2} = e^0 = 1$$

$$\left(\begin{array}{l} g(x,y) = x^2 y^2 \text{ is cont at } (0,0) \\ f(t) = e^t \text{ is cont at } 0 \end{array} \right)$$

$$\lim_{(x,y) \rightarrow (0,0)} \cos(x+y) = \cos 0 = 1$$

$$\left(\begin{array}{l} g(x,y) = x+y \text{ is cont at } (0,0) \\ f(t) = \cos t \text{ is cont at } 0 \end{array} \right)$$

$$\text{So } \lim_{(x,y) \rightarrow (0,0)} \frac{e^{x^2+y^2}}{\cos(x+y)} = \frac{1}{1} = 1$$

$$\text{Eg } \lim_{(x,y) \rightarrow (0,0)} \frac{e^{x+y} - 1}{\sin(x+y)}$$

$$g(x,y) = x+y \quad \lim_{(x,y) \rightarrow (0,0)} x+y = 0$$

$$f(t) = \frac{e^t - 1}{\sin t}$$

$$\lim_{t \rightarrow 0} f(t) = \lim_{t \rightarrow 0} \frac{e^t - 1}{\sin t} = \lim_{t \rightarrow 0} \frac{e^t}{\cos t} = 1$$

$$\text{So } \lim_{(x,y) \rightarrow (0,0)} \frac{e^{x+y} - 1}{\sin(x+y)} = 1$$

$$\text{Watch out: } \lim_{(x,y) \rightarrow (0,0)} \frac{e^{x+y} - 1}{\sin(x-y)} \quad \text{DNE}$$

pt let $x=y$ then $\sin(x-y) = \sin(0) = 0$ so

$$\frac{e^{2x} - 1}{\sin 0} \text{ DNE. let } x=-y \text{ then } \frac{e^0 - 1}{\sin(-2y)} = 0 \quad \forall y \neq 0$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2} = \text{DNE}$$

$$\begin{matrix} x=0 \\ y \neq 0 \end{matrix}$$

$$\frac{xy}{x^2+y^2} = 0$$

$$x=y \neq 0$$

$$\frac{xy}{x^2+y^2} = \frac{1}{2}$$

so limit should be 0 & $\frac{1}{2}$??

Then If $f_1(x,y) \leq f(x,y) \leq f_2(x,y)$

$$\text{and } \lim_{(x,y) \rightarrow (a,b)} f_1(x,y) = \lim_{(x,y) \rightarrow (a,b)} f_2(x,y) = L$$

$$\text{then } \lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$$

$$\text{Eg } \lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2+xy^2}{x^2+y^2} = \lim_{(x,y) \rightarrow (0,0)} 1 + \frac{xy^2}{x^2+y^2}$$

$$-|x|(x^2+y^2) \leq xy^2 \leq |x|(x^2+y^2) \quad \text{so } -|x| \leq \frac{xy^2}{x^2+y^2} \leq |x|$$

$$\lim_{(x,y) \rightarrow (0,0)} |x| = \lim_{(x,y) \rightarrow (0,0)} -|x| = 0$$

$$\text{so } \lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2+xy^2}{x^2+y^2} = 1$$

Defn $f(x, y)$ is continuous at (a, b) if

- ① $f(a, b)$ is defined
- ② $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ exists
- ③ $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = f(a, b)$

Thm If f_{xy} & f_{yx} exist and are continuous then $f_{xy} = f_{yx}$.

Ex Find the following limits or show they DNE

1) $\lim_{(x, y) \rightarrow (1, 3)} (3x^2y - xy^3)$

2) $\lim_{(x, y) \rightarrow (-1, 2)} \frac{xy - y^3}{(x + y + 1)^2}$

3) $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy + \cos x}{xy - \cos x}$

4) $\lim_{(x, y) \rightarrow (0, 0)} \frac{\tan(x^2 + y^2)}{x^2 + y^2}$

5) $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 + y^2}{x^4 - y^4}$

6) $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^4 - y^4}{x^2 + y^2}$

7) $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{(x^2 + y^2)^2}$

8) $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy(x^2 - y^2)}{x^2 + y^2}$

9) $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy^2}{x^2 + y^4}$

10) Let $f(x, y) = xy \frac{x^2 - y^2}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$ & $f(0, 0) = 0$

Show that $f(x, y)$ is continuous at $(0, 0)$ but $f_{xy}(0, 0) \neq f_{yx}(0, 0)$